

DSCI550: Data Science at Scale

Homework 5, Spring 2026

Due: 11:59 pm, April 15, 2026.

1. (20 pts) Assuming the following descriptions of a simple computer system setup:
 - The user program continuously performs read operations of 64KB data blocks which requires 2 million cycles to process each block.
 - The clock rate is 3GHz, i.e., 3 billion cycles per second.
 - The maximum sustained transfer rate of the memory bus is 640MB/sec
 - The read/write bandwidth of the hard disk drives is 64MB/sec, disk average seek time plus average rotational latency is 9ms.
 - Assume no extra overhead.
 - 1) (5 pts) How much time is needed for CPU to process each 64 KB block?
 - 2) (5 pts) How much time is spent in memory data transfer for a 64 KB block?
 - 3) (5 pts) How much time is spent in I/O transfer for a 64 KB block?
 - 4) (5 pts) Based on the answers of a) – c), what is the bottleneck (slowest component) in this computer system setup, the CPU, memory bus, or the disk set?

Use decimal system for KB (10^3) and MB (10^6).

2. (20 points) The key parameters of a hard disk are as follows.

Capacity	1TB
Rotational speed	10, 000RPM
Maximum seek time	10ms
Read Rate	50MB/sec
Block size	4KB

- 1) (10 pts) What is the average time to read 400 KB stored at randomly distributed blocks? What is the effective data transfer rate for this case?
 - 2) (10 pts) What is the average time to read 400 KB stored at sequential blocks? What is the effective data transfer rate for this case?
3. (10 pts) Briefly answer the following questions in Hadoop:
 - 1) What are the two master nodes in Hadoop? How does the master achieve high reliability?
 - 2) What is speculative execution (also called backup tasks)? What problem does it solve?
4. (20 pts) You have a data file (480MB) and want to store it in HDFS which block size is 128 MB. Your Hadoop cluster has 10 data nodes.
 - 1) Explain (draw a figure if needed) how this file is divided and stored in this system in details (e.g., with specific locations of data blocks).
 - 2) And explain the steps how this file can be accessed later.

5. (30 pts) A Hadoop cluster has 10 worker nodes, each with 4 CPU cores and 4 TB of storage. Each node can process one Map task per core in parallel.

You are given a 100 GB (1 GB = 1024 MB) dataset stored in HDFS using a 128 MB block size and a replication factor of 3.

A MapReduce job is submitted to perform simple log analysis.

Each Map task processes one data block, and the average Map task takes 90 seconds to complete.

Note that in Hadoop, a wave refers to one full round of parallel task execution — that is, the number of tasks that can be run simultaneously given the cluster's capacity. Because the number of tasks (Map or Reduce) is often much larger than the number of available task slots (CPU cores), Hadoop executes them in multiple batches. Each batch is called a wave.

Answer the following:

- 1) How many Map tasks are created for this job?
- 2) What is the *maximum* number of Map tasks that can run simultaneously on the cluster?
- 3) Assuming ideal scheduling and perfect data locality, how long will it take to complete all Map tasks?
- 4) Now suppose only 80 % of the Map tasks achieve data locality, while 20 % must fetch data remotely, adding 30 seconds of network delay per task.
Compute the new estimated completion time for all Map tasks.